

Exercise: Tar Content Analysis in Cigarettes

Lab Analysis Inc. conducted measurements to determine if the tar contents (in milligrams) differ among three different brands of cigarettes. The measurements are shown in the following table.

Sample	Brand X	Brand Y	Brand Z
1	9.95	10.82	10.70
2	10.03	10.77	10.85
3	10.10	10.89	11.00
4	9.60	10.35	11.40
5	9.58	10.45	11.30
6	9.50	10.50	11.35

Task:

1. Write the Equation of the Linear Regression Model: Assume Brand X is the baseline. Use indicator variables for Brand Y and Brand Z.
2. Estimate the Regression Coefficients: Calculate the means for each brand and the deviations of Brand Y and Brand Z from Brand X.
3. Statistical Analysis:
 - Complete the Analysis of Variance Table: List the sources of variation (Total, Regression, Residual), degrees of freedom, and deviance.
 - Homogeneity of Means Test: Perform an F-test to assess if the means of the tar content across the brands are statistically different. Use appropriate hypotheses and calculate the F-statistic.

This exercise will help understand if the differences in tar contents across these brands are significant or not. Use the provided data to carry out the linear regression analysis and complete the statistical testing.

$$N = \sum_{j=1}^3 n_j = 6 + 6 + 6 = 18$$

$$\bar{y}_1 = \frac{1}{n_1} \sum_{i=1}^6 y_{i:1} = 9.79$$

$$\bar{y}_2 = \frac{1}{n_2} \sum_{i=1}^6 y_{i:2} = 10.63$$

$$\bar{y}_3 = \frac{1}{n_3} \sum_{i=1}^6 y_{i:3} = 11.1$$

$$\bar{y} = \frac{1}{N} \sum_{j=1}^3 n_j \bar{y}_j = 10.51$$

$$Y_i = \beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i} + \epsilon_i$$

$$x_{1i} = \begin{cases} 1 & \text{if group Y} \\ 0 & \text{otherwise} \end{cases}$$

$$x_{2i} = \begin{cases} 1 & \text{if group Z} \\ 0 & \text{otherwise} \end{cases}$$

$$\mu_1 = \beta_1 = 9.79$$

$$\mu_2 = \beta_1 + \beta_2 = 10.63 \rightarrow \beta_2 = \mu_2 - \beta_1 = 0.84$$

$$\mu_3 = \beta_1 + \beta_3 = 11.1 \rightarrow \beta_3 = \mu_3 - \beta_1 = 1.31$$

$$Y_i = 9.79 + 0.84 x_{1i} + 1.31 x_{2i} + \epsilon_i$$

$$SST = \sum (y_{i:3} - \bar{y})^2$$

$$SSE = \sum \sum (y_{i:3} - \bar{y}_j)^2 = \sum n_j (y_j - \bar{y})^2$$

$$SSR = \sum (\bar{y}_j - \bar{y})^2 n_j$$